Algorithmic Authority: A Practitioner Framework for Generative Engine Optimization Based on a 7-Day Implementation Sprint

Alexandre Caramaschi Brasil GEO (Independent Researcher), Goiania, Brazil ORCID: 0009-0004-9150-485X

Corresponding author: alexandre@brasilgeo.ai

Keywords: Generative Engine Optimization, GEO, Large Language Models, Entity Consistency, Algorithmic Citability, Structured Data, Schema.org, Knowledge Graphs, B2A, Action Research

Submitted: March 2026

Abstract

The emergence of Large Language Models (LLMs) as primary information intermediaries has created an urgent need for frameworks that address how digital entities achieve visibility within AI-generated responses. This paper introduces the GEO Enterprise Framework, a 10-layer practitioner model for Generative Engine Optimization derived from action research conducted during a 7-day implementation sprint (March 17--23, 2026). The sprint produced 206 commits across 8 repositories, generated 61 pages and 92 indexable URLs, implemented 29 Schema.org types within a single JSON-LD @graph architecture, and achieved an Entity Consistency Score improvement from 20% to 80%---all at zero infrastructure cost using free-tier platforms. The framework formalizes several novel constructs: the *Invisible Excellence Paradox*, whereby technically optimized sites generate zero organic traffic without editorial authority; *Algorithmic Citability*, the structural property enabling LLM citation; and the *Entity Consistency Score* (ECS), a quantitative metric for cross-platform identity alignment. Empirical results reveal a critical gap between technical optimization and actual LLM citation, suggesting that structured data and semantic markup are necessary but insufficient conditions for algorithmic visibility. The paper contributes to the nascent GEO literature by providing both a theoretical framework and reproducible implementation data, bridging the gap between academic research and practitioner application in the post-search, agent-mediated information economy.

---

1. Introduction

The information retrieval landscape is undergoing its most fundamental transformation since the advent of web search engines in the mid-1990s. Large Language Models such as ChatGPT, Google Gemini, Perplexity, and

Claude have introduced a paradigm in which users receive synthesized, conversational answers rather than ranked lists of links [Aggarwal et al., 2023]. This shift has profound implications for how organizations, professionals, and knowledge producers achieve visibility in the digital ecosystem.

Traditional Search Engine Optimization (SEO) operated within a well-understood framework: websites competed for positions in a ranked list of ten results per page, and success was measured through click-through rates, organic traffic, and keyword rankings. The emergence of zero-click searches---where search engines provide answers directly on the results page---had already begun eroding this model. By 2024, SparkToro estimated that over 60% of Google searches ended without a click to any external website [Fishkin, 2024]. The rise of LLM-powered search accelerates this trend to its logical conclusion: the disappearance of the click as the fundamental unit of information commerce.

Generative Engine Optimization (GEO) represents the emerging discipline concerned with optimizing digital presence for inclusion in AI-generated responses. Unlike SEO, where the optimization target is a deterministic ranking algorithm, GEO must contend with probabilistic language models whose citation behavior is influenced by training data composition, retrieval-augmented generation (RAG) pipelines, and entity recognition heuristics that remain largely opaque to external practitioners.

This paper addresses a critical gap in the GEO literature. While foundational research has established that GEO techniques can improve visibility in generative engine responses by 30--40% [Aggarwal et al., 2023], there exists no comprehensive practitioner framework that integrates technical implementation, entity management, and measurement into a cohesive model. Furthermore, the existing literature lacks empirical case studies documenting full-stack GEO implementations with verifiable data.

The contributions of this paper are threefold:

1. **The GEO Enterprise Framework**: A 10-layer model that organizes the full scope of GEO activities from infrastructure through governance, providing practitioners with a structured approach to implementation.
2. **Empirical implementation data**: Quantitative results from a 7-day sprint that produced a complete GEO-optimized digital presence, with all data verifiable through public git commits and live URLs.
3. **Novel constructs**: Formalization of the *Invisible Excellence Paradox*, *Entity Consistency Score*, *Canonical Credential*, and *Algorithmic Citability* as theoretical tools for understanding and measuring GEO effectiveness.

The remainder of this paper is organized as follows. Section 2 reviews background and related work. Section 3 presents the GEO Enterprise Framework. Section 4 describes the research methodology. Section 5 reports quantitative results. Section 6 discusses findings and implications. Section 7 addresses limitations and future work. Section 8 concludes.

---

2. Background and Related Work

2.1 From SEO to GEO: The Evolution of Search Optimization

Search Engine Optimization has evolved through several distinct phases since its inception in the late 1990s. The early era (1998--2005) was characterized by keyword-centric tactics---keyword stuffing, meta tag manipulation, and link farming. Google's PageRank algorithm [Brin and Page, 1998] introduced the notion that the quality and quantity of inbound links served as a proxy for page authority, inaugurating the link-building era (2005--2012). Subsequent algorithm updates---Panda (2011), Penguin (2012), Hummingbird (2013), and RankBrain (2015)---progressively shifted the optimization target from mechanical signals to content quality, user intent, and semantic understanding.

The introduction of Knowledge Panels, Featured Snippets, and People Also Ask boxes marked the beginning of the zero-click era, in which Google itself became the destination rather than the intermediary. Fishkin and SparkToro [2024] documented that by 2024, approximately 60% of searches ended on Google's own properties without generating a single click to external websites. This represented a fundamental restructuring of the implicit contract between search engines and content producers.

The launch of ChatGPT in November 2022, followed by Google's Search Generative Experience (SGE), Perplexity AI, and integration of LLMs into Bing, introduced a qualitatively different paradigm. In generative search, there is no ranked list. The engine synthesizes a response, occasionally citing sources but frequently presenting information without attribution. The optimization target shifts from "ranking position" to "citation probability"---a fundamentally different objective with distinct technical requirements.

2.2 Foundational GEO Research

The term "Generative Engine Optimization" was formalized by Aggarwal et al. [2023] in a Princeton University study that provided the first systematic analysis of how content could be optimized for inclusion in LLM-generated responses. Their experimental framework tested nine optimization strategies across multiple generative engines and found that techniques such as citing authoritative sources, including relevant statistics, and using quotations from credible entities improved visibility by 30--40%.

This foundational study established several important principles. First, GEO and SEO are not equivalent disciplines; content that ranks well in traditional search does not necessarily receive citation in generative responses. Second, the strategies that improve GEO performance are fundamentally about *credibility signals*---indicators that help the language model assess the authoritativeness and reliability of a source. Third, the effectiveness of GEO strategies varies across domains and query types, suggesting that a one-size-fits-all approach is insufficient.

2.3 LLM Citation Behavior and Platform Distribution

A January 2026 study by Semrush analyzed citation patterns across major LLM platforms, revealing a striking distribution of source preferences. Reddit accounted for 11.29% of citations, LinkedIn for 11.03%, Wikipedia for 9.53%, and YouTube for 8.77% [Semrush, 2026]. These findings have significant implications for GEO strategy: the platforms most frequently cited by LLMs are not traditional websites but community-driven platforms where individuals establish persistent, verifiable identities.

This citation distribution suggests that LLMs have developed an implicit preference for sources that exhibit three properties: (a) community validation through engagement metrics (upvotes, comments, shares), (b) identity persistence through user profiles with verifiable history, and (c) topical consistency demonstrated through repeated engagement with specific subject areas. These properties collectively inform what this paper terms *Algorithmic Citability*.

2.4 The Zero-Click Economy and the Attention Collapse

The zero-click phenomenon has been extensively documented in the SEO literature [Fishkin, 2024; Schwartz, 2023], but its implications for the broader information economy remain underexplored. In the traditional web economy, content producers created value by attracting clicks, which were monetized through advertising, lead generation, or direct sales. The zero-click economy severs this value chain: content is consumed without the consumer ever visiting the producer's property.

LLM-mediated search extends this logic further. Not only does the user not click, but the user may not even be aware that specific sources contributed to the response they received. The content producer's work is consumed, synthesized, and delivered without attribution, traffic, or compensation. This creates what we term the *Invisible Excellence Paradox*: a state in which technically perfect, semantically rich, authoritative content generates zero measurable organic engagement despite being optimally structured for machine comprehension.

2.5 Technical Foundations: Schema.org, Knowledge Graphs, and RAG

Three technical domains provide the foundation for GEO implementation.

Schema.org and Structured Data. The Schema.org vocabulary, maintained by a W3C community group with sponsorship from Google, Microsoft, Yahoo, and Yandex, provides a shared vocabulary for marking up web content in ways that machines can understand. JSON-LD (JavaScript Object Notation for Linked Data) has emerged as the preferred serialization format, enabling webmasters to embed structured data without modifying visible HTML content [Guha et al., 2016].

Knowledge Graphs. Google's Knowledge Graph, introduced in 2012, represented one of the first large-scale efforts to organize web information into a graph of entities and relationships [Singhal, 2012]. In the context of GEO, knowledge graphs serve as the substrate upon which LLMs verify entity identity, resolve ambiguity, and assess authority. An entity that exists as a well-connected node in the knowledge graph---with consistent properties, verified relationships, and multiple corroborating sources---is more likely to be recognized, trusted, and cited by generative engines.

Retrieval-Augmented Generation (RAG). RAG architectures [Lewis et al., 2020] augment language model generation with retrieved passages from external knowledge bases. In RAG-based search systems (e.g., Perplexity, Bing Chat), the model first retrieves relevant documents and then generates a response conditioned on both the query and the retrieved context. This architecture creates a direct path from structured, accessible content to LLM citation, making technical accessibility and semantic clarity direct determinants of visibility.

2.6 The llms.txt Specification

The llms.txt specification, proposed by Jeremy Howard [2024], introduces a standardized mechanism for websites to provide machine-readable content specifically designed for LLM consumption. Analogous to robots.txt for search engine crawlers, llms.txt provides a curated, structured representation of a site's content that LLMs can efficiently parse and incorporate into their knowledge bases. The specification represents an explicit acknowledgment that the audience for web content now includes machines as well as humans, and that these audiences have different information needs and consumption patterns.

---

3. The GEO Enterprise Framework

3.1 Framework Overview

The GEO Enterprise Framework is a 10-layer model that organizes the full scope of Generative Engine Optimization activities into a hierarchical stack. Each layer builds upon the capabilities established by lower layers, creating a progressive implementation path from foundational infrastructure through strategic governance.

The framework emerged inductively from the implementation sprint documented in this paper. Rather than being designed *a priori* and then implemented, the layers were identified through systematic reflection on the activities, dependencies, and outcomes observed during the sprint. This grounded, practice-based origin distinguishes the framework from purely theoretical models.

Table 1. The GEO Enterprise Framework: 10-Layer Model

Layer	Name	Function	Key Metrics
L1	Infrastructure	Hosting, domains, CDN, performance	TTFB, LCP, uptime
L2	Structured Data	Schema.org markup, JSON-LD, knowledge graph signals	Schema types, validation errors
L3	Conversion Tracking	Event capture, funnel measurement	Tracking points, event coverage

Layer	Name	Function	Key Metrics
L4	Measurement	Analytics, dashboards, reporting	Data sources, metric coverage
L5	Automation	CI/CD, scheduled tasks, data pipelines	Automated workflows, update frequency
L6	Editorial	Content creation, cross-platform publishing	Articles published, platform presence
L7	Entity Consistency	Cross-platform identity alignment	Entity Consistency Score (ECS)
L8	Ads Preparation	Paid amplification readiness	Pixel installation, audience definitions
L9	Academic Publishing	Research output, citation generation	Papers published, citations received
L10	Governance	Strategy, compliance, knowledge management	Documentation coverage, decision log

3.2 Layer 1: Infrastructure

The infrastructure layer establishes the technical foundation upon which all subsequent GEO activities depend. In the context of the implementation sprint, this layer was realized through a dual-platform architecture: a Next.js application deployed on Vercel (alexandrecaramaschi.com) and a Cloudflare Pages site (brasilgeo.ai), both operating on free-tier plans at zero monthly cost.

Key infrastructure decisions with GEO implications include:

- **Edge deployment**: Both Vercel and Cloudflare deploy to global edge networks, ensuring low Time-to-First-Byte (TTFB) regardless of user or crawler location.
- **Static generation**: Pre-rendered pages ensure that content is immediately available to crawlers without requiring JavaScript execution, a critical factor for both search engine and LLM accessibility.
- **Zero-cost architecture**: The demonstration that enterprise-grade GEO infrastructure can be built at zero cost has implications for accessibility and democratization of the discipline.

3.3 Layer 2: Structured Data

The structured data layer implements the semantic markup that enables machines to understand the identity, relationships, and authority signals embedded in web content. The sprint implementation deployed 29 Schema.org types within a single JSON-LD @graph, creating a dense, interconnected representation of the entity and its activities.

The @graph architecture is significant because it establishes explicit relationships between entities rather than presenting them as isolated fragments. A Person entity is connected to an Organization through the `worksFor` property, which is connected to WebPage entities through `mainEntity`, which reference Article entities through `hasPart`. This interconnected graph mirrors the structure of knowledge graphs maintained by search engines and LLM providers.

Table 2. Schema.org Types Implemented in the Sprint

Category	Types	Count
Identity	Person, Organization, Brand	3
Content	Article, BlogPosting, WebPage, WebSite, FAQPage	5
Professional	ProfessionalService, Offer, Service, Product	4
Events	Event, Course, EducationEvent	3
Media	VideoObject, ImageObject, MediaObject	3
Navigation	BreadcrumbList, SiteNavigationElement, ItemList	3
Metadata	SearchAction, ContactPoint, PostalAddress, GeoCoordinates	4
Authority	Credential, ScholarlyArticle, Book, HowTo	4

3.4 Layer 3: Conversion Tracking

While GEO is primarily concerned with visibility rather than direct response, the conversion tracking layer ensures that any traffic or engagement generated by LLM citations can be measured and attributed. The sprint implementation deployed 12 distinct conversion tracking points across multiple interaction types (WhatsApp clicks, form submissions, content downloads, page depth engagement, scroll milestones, and external link exits).

3.5 Layer 4: Measurement

The measurement layer establishes the data collection and analysis infrastructure required to quantify GEO effectiveness. The sprint implementation integrated 9 automated data sources, including Google Search Console, web analytics, uptime monitoring, and custom Schema.org validation tools. Eight statistical functions were developed for temporal analysis, enabling trend detection and anomaly identification across GEO metrics.

3.6 Layer 5: Automation

The automation layer reduces manual effort and ensures consistency through programmatic workflows. In the sprint, automation was applied to sitemap generation, IndexNow submission, structured data validation, content deployment, and cross-platform publishing. The 8 repositories used in the sprint were managed through automated CI/CD pipelines that ensured consistency between development and production environments.

3.7 Layer 6: Editorial

The editorial layer addresses the creation and distribution of content across platforms that LLMs preferentially cite. Informed by the Semrush [2026] citation distribution data, the sprint's editorial strategy targeted LinkedIn (11.03% of LLM citations), YouTube (8.77%), Reddit, Medium, Hashnode, and Quora. The objective is to create a constellation of topically consistent, authoritative content nodes across the platforms that constitute the LLM's training and retrieval corpus.

3.8 Layer 7: Entity Consistency

Entity Consistency is arguably the most distinctive contribution of the GEO Enterprise Framework. This layer addresses the alignment of identity representations across all digital platforms where the entity (person, organization, or brand) maintains a presence.

The *Entity Consistency Score* (ECS) is defined as:

$$ECS = (1/N) * \Sigma_{i=1}^{N} C_i$$

Where N is the number of platforms assessed, and C_i is the consistency score for platform i, measured across five dimensions:

1. **Name consistency** (0--1): Exact match of canonical name across platforms.
2. **Credential consistency** (0--1): Use of the *Canonical Credential*---the single authoritative description---across all bios and profiles.
3. **Visual consistency** (0--1): Use of the same professional photograph and brand assets.
4. **Topical consistency** (0--1): Alignment of content themes and expertise claims.
5. **Link consistency** (0--1): Correct and consistent URLs pointing to canonical properties.

During the sprint, the ECS improved from 20% on Day 1 to 50% by Day 4, and subsequently reached 80% through continued editorial activity. The initial 20% score reflected a common state: the entity existed across multiple platforms but with inconsistent names, outdated descriptions, conflicting credentials, and broken links.

The *Canonical Credential* is a carefully constructed, standardized identity description designed to be used verbatim across all platforms. For the sprint subject, the Canonical Credential defined the exact professional title, organizational affiliation, notable achievements, and areas of expertise in a format optimized for both human readability and machine parsing.

3.9 Layer 8: Ads Preparation

The ads preparation layer establishes the infrastructure for paid amplification when organic GEO channels prove insufficient. This includes pixel installation (Meta, Google, LinkedIn), audience definition, and campaign structure planning. While the sprint operated at zero cost, this layer acknowledges that the *Invisible Excellence Paradox* may necessitate paid distribution to achieve the initial visibility threshold required for organic citation.

3.10 Layer 9: Academic Publishing

The academic publishing layer leverages the unique authority signal that peer-reviewed and scholarly publications provide within LLM training corpora. Academic papers indexed in Google Scholar, arXiv, ResearchGate, and similar platforms carry implicit authority signals that influence citation probability. This paper itself represents an output of Layer 9.

3.11 Layer 10: Governance

The governance layer provides the strategic oversight, documentation, and decision-making framework that ensures coherent, sustainable GEO activity. During the sprint, governance was maintained through comprehensive documentation of decisions, rationales, and outcomes, including a chronological evolution log and a consolidated governance document.

---

4. Methodology

4.1 Research Approach

This study employs *action research* [Lewin, 1946; Susman and Evered, 1978], a methodology in which the researcher is simultaneously the implementer and the observer. Action research is particularly appropriate for GEO investigation because the discipline is too nascent for controlled experiments---there are no established benchmarks, standardized metrics, or control conditions against which to test. The researcher's dual role as practitioner and analyst enables the capture of implementation nuances, decision rationales, and emergent insights that would be inaccessible to external observers.

4.2 Implementation Context

The implementation sprint was conducted between March 17 and March 23, 2026 (7 calendar days, with intensive development concentrated in the final 5 days). The subject entity was a GEO practitioner and consultant seeking to establish algorithmic authority across generative engines.

The sprint began from a near-zero baseline: existing websites with minimal structured data, inconsistent cross-platform identity, no llms.txt implementation, limited Schema.org markup, and no systematic measurement infrastructure. This starting condition enabled documentation of the full implementation trajectory.

4.3 Technical Environment

- **Primary framework**: Next.js 14 (React-based)
- **Hosting**: Vercel (free tier) and Cloudflare Pages (free tier)
- **Version control**: Git, with 8 repositories across GitHub
- **Structured data**: JSON-LD with Schema.org vocabulary
- **Development tools**: Claude Code (AI-assisted development), VS Code
- **Analytics**: Custom dashboards with automated data collection
- **Monitoring**: Uptime and performance monitoring via free-tier services

4.4 Data Collection

Data was collected through multiple channels:

1. **Git commit logs**: Providing timestamped, granular records of all implementation activities (206 commits over 5 days).
2. **Google Search Console**: Tracking indexation status, crawl activity, and search impressions.
3. **Web analytics**: Recording traffic volume, sources, and engagement patterns.
4. **Manual platform audits**: Assessing Entity Consistency Score across targeted platforms.
5. **Schema.org validation**: Automated testing of structured data accuracy and completeness.
6. **LLM query testing**: Systematic querying of ChatGPT, Gemini, Perplexity, and Claude to assess citation behavior.

4.5 Measurement Dimensions

The sprint was assessed across five measurement dimensions:

Table 3. Measurement Dimensions and Indicators

Dimension	Indicators	Data Source
Development velocity	Commits, pages created, components built	Git logs
Technical completeness	Schema types, validation status, performance scores	Automated validation
Entity consistency	ECS across platforms	Manual audit
Indexation progress	URLs indexed, crawl frequency	Google Search Console
Visibility outcomes	Organic traffic, LLM citations	Analytics, LLM testing

---

5. Results

5.1 Development Velocity

The sprint produced implementation output at an intensity that would be extraordinary for a solo practitioner operating without AI-assisted development tools. Table 4 summarizes the quantitative outputs.

Table 4. Sprint Output Summary (March 17--23, 2026)

Metric	Value
Total commits	206
Active development days	5 (March 19--23)
Average commits per day	41.2
Pages created	61
React components built	35
Indexable URLs generated	92 (later 97)
Repositories used	8
Schema.org types implemented	29
Conversion tracking points	12
Automated data sources	9
Statistical analysis functions	8
llms.txt version reached	v9.0 (68+ links)
Infrastructure cost	$0/month

The 206 commits across 5 days represent an average of 41.2 commits per day, or approximately one commit every 11.6 minutes during a 8-hour workday. This velocity was enabled by AI-assisted development using Claude Code, which handled code generation, debugging, and architectural decisions, allowing the practitioner to focus on strategic direction and content decisions.

5.2 Structured Data Implementation

The structured data implementation represented one of the sprint's most technically ambitious outputs. Twenty-nine Schema.org types were integrated into a single JSON-LD @graph, creating a densely interconnected knowledge representation.

Figure 1. Schema.org @graph Architecture (Simplified)

```
@graph
├── WebSite
```

```
│   ├── SearchAction
│   └── publisher → Organization
├── Person (mainEntity)
│   ├── worksFor → Organization
│   ├── alumniOf → [Organizations]
│   ├── knowsAbout → [Topics]
│   ├── hasCredential → [Credentials]
│   └── sameAs → [Platform URLs]
├── Organization
│   ├── founder → Person
│   ├── offers → [Services]
│   └── contactPoint → ContactPoint
├── WebPage (×61)
│   ├── mainEntity → varies
│   ├── breadcrumb → BreadcrumbList
│   └── speakable → SpeakableSpecification
├── Article / BlogPosting (×N)
│   ├── author → Person
│   └── publisher → Organization
├── FAQPage
│   └── mainEntity → [Question/Answer]
├── Course / Event
│   ├── instructor → Person
│   └── organizer → Organization
└── VideoObject (×N)
    ├── author → Person
    └── thumbnailUrl → ImageObject
```

The @graph structure ensures that entities are defined once and referenced throughout, eliminating the inconsistencies that arise when Schema.org types are implemented as isolated fragments across different pages.

5.3 Entity Consistency Score Progression

The Entity Consistency Score was measured at four time points during the sprint, yielding the trajectory shown in Table 5.

Table 5. Entity Consistency Score Progression

Time Point	ECS	Key Changes
Day 1 (March 17)	20%	Baseline: inconsistent names, outdated bios, broken links
Day 3 (March 19)	35%	Canonical Credential drafted; primary platforms updated
Day 5 (March 21)	50%	LinkedIn, GitHub, YouTube aligned; Schema.org Person entity deployed

Time Point	ECS	Key Changes
Day 7+ (March 23+)	80%	Secondary platforms updated; visual assets standardized

The progression from 20% to 80% demonstrates that Entity Consistency is achievable through systematic, focused effort but requires attention to a long tail of platforms. The initial rapid improvement (20% to 50%) came from aligning the highest-impact platforms (LinkedIn, GitHub, personal website). The subsequent improvement (50% to 80%) required updating secondary platforms (Medium, Hashnode, Twitter/X, Quora, Reddit) where the entity maintained dormant or inconsistent profiles.

5.4 Indexation and Crawl Activity

By the end of the sprint, 92 URLs had been submitted for indexation (later growing to 97 as additional content was published). Google Search Console data indicated progressive crawling activity, with the indexation pipeline processing new URLs within 24--48 hours of submission via IndexNow.

5.5 The Visibility Paradox: Zero Organic Traffic

Despite the technical completeness documented above---29 Schema.org types, 92 indexable URLs, 61 pages of content, a comprehensive llms.txt file, and an Entity Consistency Score of 80%---the implementation generated zero organic search traffic during the measurement period.

This finding, initially counterintuitive, constitutes one of the paper's most significant empirical contributions. We term this the *Invisible Excellence Paradox*.

Table 6. Expected vs. Observed Outcomes

Metric	Expected	Observed	Gap
Organic search traffic	> 0	0	Total
LLM citations (ChatGPT)	Some	0	Total
LLM citations (Perplexity)	Some	0	Total
LLM citations (Gemini)	Some	0	Total
Schema.org validation	Pass	Pass	None
PageSpeed score	> 90	> 90	None
Indexation requests	Submitted	Accepted	None

The paradox is clear: the implementation was technically flawless by every measurable standard, yet it produced no visibility outcome. This finding has profound implications for the GEO discipline, discussed in Section 6.

6. Discussion

6.1 The Invisible Excellence Paradox

The central finding of this study---that a technically perfect GEO implementation generates zero organic visibility---challenges the implicit assumption in both the SEO and GEO literature that technical optimization is sufficient for visibility. We propose that the Invisible Excellence Paradox arises from a misunderstanding of the relationship between *technical readiness* and *editorial authority*.

Technical optimization (structured data, performance, accessibility, semantic markup) creates the *capacity* for algorithmic citation. It makes content *citable*---parseable, verifiable, and semantically rich. However, it does not create the *motivation* for citation. LLMs cite sources because those sources appear in their training data, retrieval indices, or knowledge graphs with sufficient frequency, consistency, and authority to be selected during the generation process.

We propose a two-factor model of algorithmic visibility:

```
Algorithmic Visibility = f(Technical Readiness × Editorial Authority)
```

Where: - **Technical Readiness** encompasses structured data, performance, accessibility, Schema.org markup, llms.txt, and similar implementation factors. - **Editorial Authority** encompasses content volume, citation frequency in LLM training data, cross-platform presence, community engagement, and temporal persistence.

The multiplicative relationship is critical: if either factor is zero, visibility is zero regardless of the other factor's magnitude. The sprint achieved near-perfect Technical Readiness but had insufficient Editorial Authority (no existing corpus of cited content, minimal cross-platform publishing history, no academic citation record), resulting in zero visibility.

This model explains a phenomenon frequently observed in practice: established entities with poor technical implementation (no Schema.org, slow sites, no llms.txt) receive LLM citations because their Editorial Authority is high (extensive publishing history, frequent citation in training data, strong platform presence). Conversely, new entities with perfect technical implementation receive no citations because they lack Editorial Authority entirely.

6.2 Implications for the Princeton GEO Study

The Aggarwal et al. [2023] finding that GEO techniques improve visibility by 30--40% should be interpreted carefully in light of the Invisible Excellence Paradox. The Princeton study measured improvement for content that was *already within* the generative engine's retrieval scope. A 30--40% improvement over a baseline of zero remains zero. The study's findings are valid for established entities seeking to optimize their existing visibility, but they do not address the more fundamental challenge of achieving *initial* visibility---what we might call the "cold start problem" of GEO.

6.3 Algorithmic Citability as a Structural Property

We define *Algorithmic Citability* as the property of digital content being structured such that a Large Language Model can identify, attribute, and cite it within a generated response. Citability is a necessary but not sufficient condition for citation. A citable source may never be cited if it does not exist within the LLM's training data or retrieval index.

Algorithmic Citability has four dimensions:

1. **Structural citability**: The content is marked up with structured data that enables entity resolution (Schema.org, JSON-LD).
2. **Semantic citability**: The content uses clear, unambiguous language with defined entities, claims, and relationships.
3. **Accessibility citability**: The content is technically accessible to crawlers and retrieval systems (fast loading, clean HTML, llms.txt).
4. **Authority citability**: The content includes signals that enable the LLM to assess trustworthiness (credentials, affiliations, citations to authoritative sources).

The sprint achieved high scores across all four dimensions. The failure to achieve actual citation suggests a fifth, implicit dimension: **corpus presence**---the extent to which the content or its source exists within the LLM's existing knowledge base.

6.4 The B2A Paradigm

The implementation sprint provides early evidence for what we term the *Business-to-Agent* (B2A) paradigm: a business model in which the primary audience for digital content is not human users but AI agents that mediate information discovery and transaction initiation.

In the B2A paradigm, the optimization target shifts from human attention metrics (page views, time on site, bounce rate) to agent interaction metrics (citation frequency, entity recognition accuracy, recommendation probability). The GEO Enterprise Framework represents an early attempt to operationalize this paradigm shift.

The B2A concept extends existing models (B2C, B2B) by recognizing that AI agents increasingly function as autonomous intermediaries. When a user asks an AI assistant to "find a GEO consultant in Brazil," the agent's response is not a list of search results but a recommendation---and the factors that influence that recommendation (entity consistency, structured data, editorial presence) are the factors addressed by the GEO Enterprise Framework.

6.5 Entity Consistency as a Knowledge Graph Signal

The improvement in Entity Consistency Score from 20% to 80% represents more than a cosmetic alignment exercise. Cross-platform consistency creates what knowledge graph researchers term *corroborating evidence*---multiple independent sources that confirm the same facts about an entity [Dong et al., 2014].

When a Language Model encounters the same name, title, affiliation, and expertise description across LinkedIn, GitHub, a personal website, YouTube, and academic publications, it can resolve the entity with higher confidence and assign it greater authority. Conversely, inconsistent representations (different names, conflicting titles, outdated affiliations) create entity ambiguity that reduces the model's confidence and, consequently, citation probability.

The Entity Consistency Score provides practitioners with a quantifiable metric for an optimization dimension that has previously been addressed through intuition or anecdote.

6.6 The Role of AI-Assisted Development in GEO

The sprint's development velocity---206 commits in 5 days by a solo practitioner---was enabled by AI-assisted development using Claude Code. This velocity has implications for the accessibility and scalability of GEO. If comprehensive GEO implementation can be achieved by a single practitioner in a week using AI tools at zero infrastructure cost, the barrier to entry for GEO is significantly lower than for traditional SEO, which typically required months of work by specialized teams.

However, this accessibility creates a paradox of its own: if every entity can achieve technical GEO perfection easily, the differentiating factor becomes Editorial Authority---the one dimension that cannot be compressed, automated, or instantly generated. This suggests that the long-term equilibrium of GEO may resemble academic publishing more than SEO: a discipline where authority is built through sustained, high-quality contribution over time, and where technical competence is merely table stakes.

---

7. Limitations and Future Work

7.1 Limitations

This study has several important limitations that constrain the generalizability of its findings.

Single-case design. The study documents a single implementation sprint for a single entity in a single domain (GEO consulting). While the framework is designed to be generalizable, empirical validation across multiple entities, domains, and contexts is needed.

Short measurement window. The sprint covered 7 days of implementation and immediate post-implementation measurement. GEO effects may take weeks or months to materialize as content is crawled, indexed, incorporated into training data, and reflected in LLM responses. The zero-traffic finding may be a timing artifact rather than a structural limitation.

Practitioner-researcher identity. The action research methodology, while appropriate for nascent fields, introduces the risk of confirmation bias. The researcher's dual role as implementer and evaluator creates inherent conflicts that must be acknowledged.

Opacity of LLM citation mechanisms. The internal workings of LLM citation selection remain opaque, making it impossible to definitively attribute citation or non-citation to specific factors. The causal claims in this paper are necessarily tentative.

Entity Consistency Score validation. The ECS metric proposed in this paper has not been independently validated. The five dimensions and their weighting (equal in the current formulation) require empirical testing to determine optimal calibration.

Free-tier constraints. While the zero-cost architecture demonstrates accessibility, free-tier platforms impose limitations (bandwidth caps, build time limits, feature restrictions) that may not be acceptable for high-traffic implementations.

7.2 Future Work

Several research directions emerge from this study:

1. **Longitudinal measurement.** Extending the measurement window to 3, 6, and 12 months to capture delayed GEO effects and test whether the Invisible Excellence Paradox resolves over time as editorial authority accumulates.
2. **Multi-entity replication.** Applying the GEO Enterprise Framework to entities across different domains (healthcare, legal, financial services, technology) to test generalizability.
3. **ECS validation.** Conducting a large-scale study to validate the Entity Consistency Score against actual LLM citation outcomes, potentially identifying which consistency dimensions have the greatest impact.
4. **Editorial Authority measurement.** Developing a quantitative metric for Editorial Authority analogous to the ECS, enabling the two-factor visibility model to be empirically tested.
5. **LLM citation tracking.** Building automated systems that systematically query LLMs and track citation behavior over time, creating a longitudinal dataset of citation patterns.
6. **Competitive analysis.** Comparing entities that receive LLM citations against those that do not, controlling for technical implementation quality, to isolate the factors that distinguish cited from uncited entities.

7. **B2A economics.** Investigating the economic implications of the B2A paradigm, including the value of an LLM citation relative to a traditional search click, and the optimal investment allocation between Technical Readiness and Editorial Authority.

---

8. Conclusion

This paper has presented the GEO Enterprise Framework, a 10-layer practitioner model for Generative Engine Optimization derived from a 7-day implementation sprint that produced 206 commits, 61 pages, 29 Schema.org types, and 92 indexable URLs at zero infrastructure cost. The framework formalizes several novel constructs---the Invisible Excellence Paradox, Entity Consistency Score, Canonical Credential, and Algorithmic Citability---that provide both theoretical and practical tools for navigating the transition from search-mediated to AI-mediated information discovery.

The study's most counterintuitive finding---that technically perfect implementation generated zero organic visibility---challenges the assumption that technical optimization is sufficient for GEO success. We propose that algorithmic visibility is a multiplicative function of Technical Readiness and Editorial Authority, and that either factor being zero produces zero visibility regardless of the other's magnitude. This finding has immediate practical implications: practitioners should allocate effort between technical implementation and editorial presence rather than pursuing technical perfection in isolation.

The GEO Enterprise Framework provides a structured approach to this balanced allocation. Its 10 layers---from Infrastructure through Governance---encompass both technical and editorial dimensions, acknowledging that GEO is a holistic discipline requiring expertise in web development, semantic technology, content strategy, identity management, and measurement science.

As LLMs increasingly mediate information discovery, the question of how entities achieve algorithmic visibility will grow in economic and social significance. The framework, constructs, and empirical findings presented in this paper represent an early contribution to a discipline that will demand sustained research attention in the years ahead. The transition from Business-to-Consumer and Business-to-Business to Business-to-Agent is not a distant speculation; it is an unfolding reality documented in the data of this sprint.

The Invisible Excellence Paradox is not a failure---it is a finding. It reveals that the rules of algorithmic visibility are fundamentally different from the rules of search visibility, and that understanding these new rules requires new frameworks, new metrics, and new research. This paper has attempted to provide all three.

References

Aggarwal, P., Murahari, V., Rajpurohit, T., Kalyan, A., Narasimhan, K., & Deshpande, A. (2023). GEO: Generative Engine Optimization. *arXiv preprint arXiv:2311.09735*. Princeton University.

Brin, S., & Page, L. (1998). The anatomy of a large-scale hypertextual web search engine. *Computer Networks and ISDN Systems*, 30(1--7), 107--117.

Dong, X. L., Gabrilovich, E., Heitz, G., Horn, W., Lao, N., Murphy, K., ... & Zhang, W. (2014). Knowledge vault: A web-scale approach to probabilistic knowledge fusion. *Proceedings of the 20th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 601--610.

Fishkin, R. (2024). Zero-click search study: How much traffic does Google keep for itself? *SparkToro Research Reports*. Retrieved from https://sparktoro.com/blog/

Guha, R. V., Brickley, D., & Macbeth, S. (2016). Schema.org: Evolution of structured data on the web. *Communications of the ACM*, 59(2), 44--51.

Howard, J. (2024). llms.txt: A proposal for standardized LLM-readable website content. Retrieved from https://llmstxt.org/

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., ... & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems*, 33, 9459--9474.

Lewin, K. (1946). Action research and minority problems. *Journal of Social Issues*, 2(4), 34--46.

Schwartz, B. (2023). Google search results increasingly dominated by zero-click answers. *Search Engine Roundtable*. Retrieved from https://www.seroundtable.com/

Semrush. (2026). LLM citation study: Which platforms do AI models cite most frequently? *Semrush Research*, January 2026. Retrieved from https://www.semrush.com/blog/

Singhal, A. (2012). Introducing the Knowledge Graph: Things, not strings. *Google Official Blog*. Retrieved from https://blog.google/

Susman, G. I., & Evered, R. D. (1978). An assessment of the scientific merits of action research. *Administrative Science Quarterly*, 23(4), 582--603.

W3C. (2023). Schema.org community group. World Wide Web Consortium. Retrieved from https://www.w3.org/community/schemaorg/

Data Availability Statement: All implementation data referenced in this paper is verifiable through public GitHub repositories and live URLs at alexandrecaramaschi.com and brasilgeo.ai. Git commit histories provide timestamped evidence for all quantitative claims regarding development velocity and technical implementation.

Conflict of Interest: The author is the founder and CEO of Brasil GEO, the organization whose GEO implementation is documented in this study. This dual role as practitioner and researcher is inherent to the action research methodology employed but represents a potential conflict that readers should consider when evaluating the findings.

Funding: This research received no external funding. All infrastructure costs were zero (free-tier platforms).

Manuscript word count: approximately 6,200 words (excluding tables, figures, and references).